Vaishnavi Rajguru- 23101A2004

Experiment 2: Automating Version Control and Code Collaboration of TE Mini Project with Git

Aim: To Perform Code Collaboration of TE Mini Project using GIT workflow

Problem:

TechCo's development team is collaborating on a new feature for their application. However, multiple developers are working on the same parts of the codebase, causing merge conflicts, inconsistent commits, and errors in version control. The team struggles to manage branches effectively and maintain a clean history. The current workflow relies heavily on manual processes, making it difficult to track and resolve issues efficiently.

Implementation:

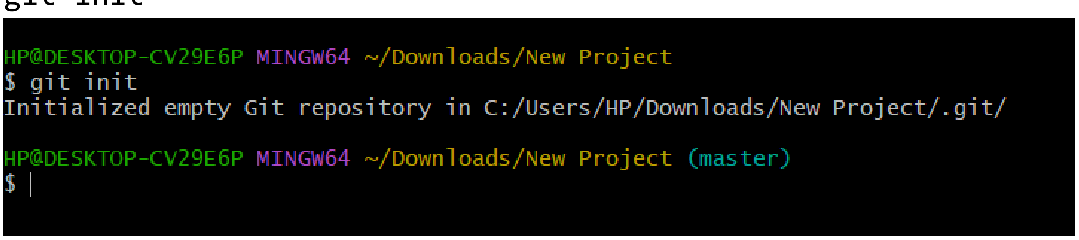
- Set up a Git repository for a project (either create a new project or use an existing one).
- Collaborate with a team member by creating multiple branches for different features.
- Implement the following:
  - Create a feature branch and make changes to the code.
  - Push changes to the remote repository.
  - Use a Git cheat sheet to commit, merge branches, and resolve conflicts.
  - Create a pull request (PR) and ensure code review processes are followed.

Step 1: Initialize a Git Repository and Push It to GitHub

1.1 Initialize a Git Repository Locally

- Navigate to your project directory on your local machine (or create a new project folder).
- Open your terminal and run the following command to initialize a Git repository:

  `git init`



1.2 Add Files to the Repository

- Add a new file or make changes to existing files. For example, create a `README.md` file in the project folder:

  `echo "# My Project" > README.md`

```
HP@DESKTOP-CV29E6P MINGW64 ~/Downloads/New Project (master)
$ echo "#My Project"> README.md

HP@DESKTOP-CV29E6P MINGW64 ~/Downloads/New Project (master)
$ echo "<html><body>Welcome</body></html>" > index.html
```

1.3 Stage and Commit Files

- Stage the files for commit using:

  `git add .`

```
HP@DESKTOP-CV29E6P MINGW64 ~/Downloads/New Project (master)
$ git add .
warning: in the working copy of 'README.md', LF will be replac
ed by CRLF the next time Git touches it
warning: in the working copy of 'index.html', LF will be repla
ced by CRLF the next time Git touches it
```

- Commit the staged files with a message:

  git commit -m "Initial commit"

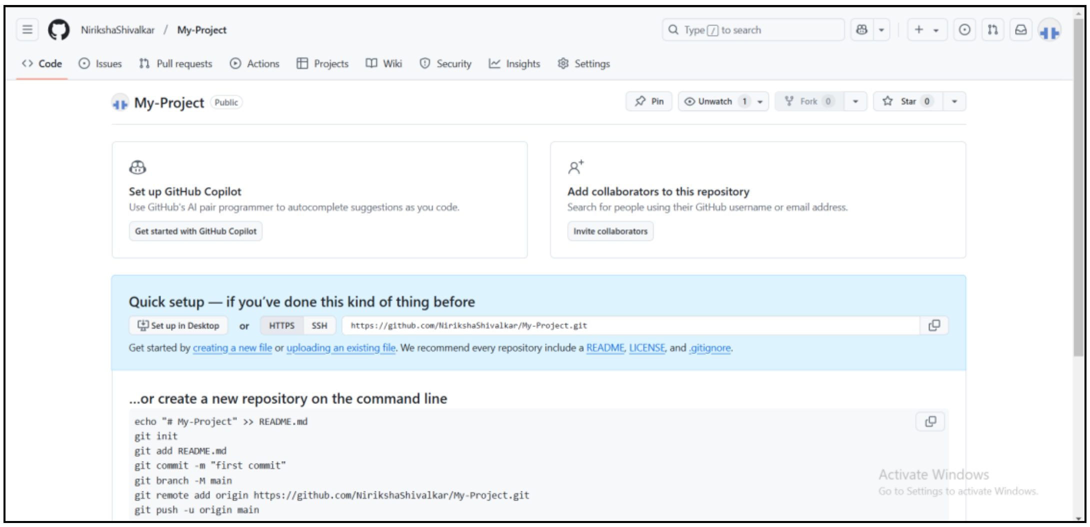
```
HP@DESKTOP-CV29E6P MINGW64 ~/Downloads/New Project (master)
$ git config --global user.email "nirikshshivalkar@gmail.com"

HP@DESKTOP-CV29E6P MINGW64 ~/Downloads/New Project (master)
$ git config --global user.name "Niriksha Shivalkar"

HP@DESKTOP-CV29E6P MINGW64 ~/Downloads/New Project (master)
$ git commit -m "Initial commit"
[master (root-commit) 6a0b64f] Initial commit
 2 files changed, 2 insertions(+)
 create mode 100644 README.md
 create mode 100644 index.html
```

1.4 Create a GitHub Repository

- Go to [GitHub](https://github.com) and create a new repository. You can name it something like "my-project".
- Copy the URL of the GitHub repository, e.g., `https://github.com/your-username/my-project.git`.



1.5 Push to GitHub

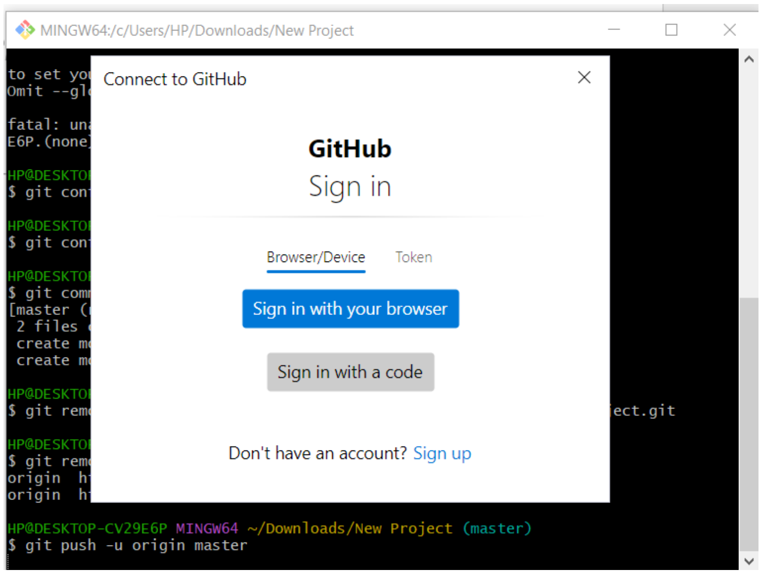
- Set the remote repository URL:

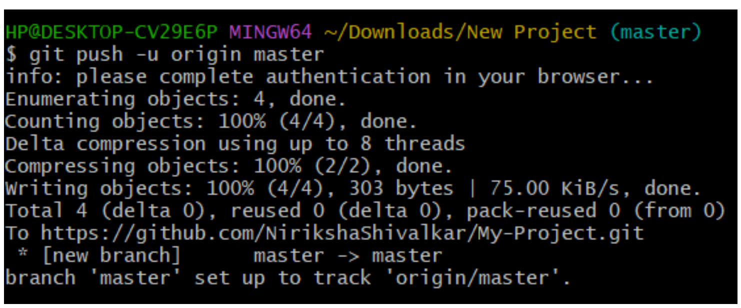
  `git remote add origin https://github.com/your-username/my-project.git`

```
HP@DESKTOP-CV29E6P MINGW64 ~/Downloads/New Project (master)
$ git remote add origin https://github.com/NirikshaShivalkar/My-Project.git

HP@DESKTOP-CV29E6P MINGW64 ~/Downloads/New Project (master)
$ git remote -v
origin  https://github.com/NirikshaShivalkar/My-Project.git (fetch)
origin  https://github.com/NirikshaShivalkar/My-Project.git (push)
```

- Push the local repository to GitHub:
  `git push -u origin master`

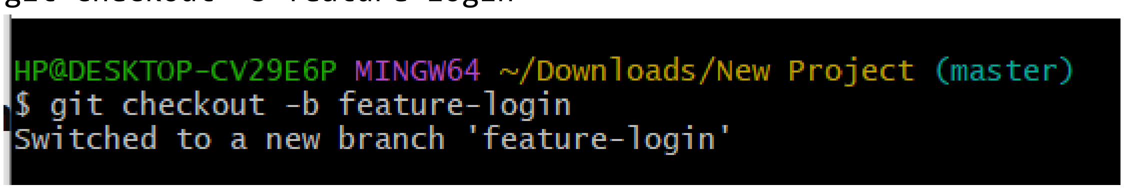




Step 2: Create and Merge Multiple Feature Branches

2.1 Create a New Feature Branch

- Create a new branch for a feature. For example, create a `feature-login` branch:
  `git checkout -b feature-login`



2.2 Make Changes in the Feature Branch

- Add a new file or modify an existing file in the `feature-login` branch. For example, modify `README.md`:

```
echo "Login feature" >> README.md
```

```
HP@DESKTOP-CV29E6P MINGW64 ~/Downloads/New Project (feature-login)
$ echo "Login feature" >> README.md
```

2.3 Stage and Commit Changes

- Stage the changes and commit them:

```
git add README.md
git commit -m "Add login feature to README"
```

```
HP@DESKTOP-CV29E6P MINGW64 ~/Downloads/New Project (feature-login)
$ git add README.md
warning: in the working copy of 'README.md', LF will be replaced by CRLF the nex
t time Git touches it

HP@DESKTOP-CV29E6P MINGW64 ~/Downloads/New Project (feature-login)
$ git commit -m "Add login  feature to README"
[feature-login a060d78] Add login  feature to README
 1 file changed, 1 insertion(+)
```

2.4 Push the Feature Branch to GitHub

- Push the feature branch to GitHub:

```
git push origin feature-login
```

```
HP@DESKTOP-CV29E6P MINGW64 ~/Downloads/New Project (feature-login)
$ git push origin feature-login
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 330 bytes | 330.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature-login' on GitHub by visiting:
remote:      https://github.com/NirikshaShivalkar/My-Project/pull/new/feature-log
in
remote:
To https://github.com/NirikshaShivalkar/My-Project.git
 * [new branch]      feature-login -> feature-login
```

2.5 Create Another Feature Branch

- Create another branch for a different feature, for example, `feature-registration`:
- `git checkout -b feature-registration`

```
HP@DESKTOP-CV29E6P MINGW64 ~/Downloads/New Project (feature-login)
$ git checkout -b feature-registration
Switched to a new branch 'feature-registration'
```

2.6 Make Changes in the `feature-registration` Branch

- Modify the same file (`README.md`), but with content related to registration:

  `echo "Registration feature" >> README.md`

```
HP@DESKTOP-CV29E6P MINGW64 ~/Downloads/New Project (feature-registration)
$ echo "Registration Feature" >> README.md
```

2.7 Stage and Commit Changes

- Stage and commit the changes:

  `git add README.md`

  `git commit -m "Add registration feature to README"`

```
HP@DESKTOP-CV29E6P MINGW64 ~/Downloads/New Project (feature-registration)
$ git add README.md
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next
 time Git touches it

HP@DESKTOP-CV29E6P MINGW64 ~/Downloads/New Project (feature-registration)
$ git commit -m "Add registration feature to README"
[feature-registration 8fe14f0] Add registration feature to README
 1 file changed, 1 insertion(+)
```

2.8 Push the `feature-registration` Branch to GitHub

- Push the registration branch to GitHub:

  `git push origin feature-registration`

```
HP@DESKTOP-CV29E6P MINGW64 ~/Downloads/New Project (feature-registration)
$ git push origin feature-registration
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 349 bytes | 174.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature-registration' on GitHub by visiting:
remote:      https://github.com/NirikshaShivalkar/My-Project/pull/new/feature-reg
istration
remote:
To https://github.com/NirikshaShivalkar/My-Project.git
 * [new branch]      feature-registration -> feature-registration
```

Step 3: Simulate a Merge Conflict and Resolve It

3.1 Switch to `master` Branch

- Now switch back to the `master` branch:

  `git checkout master`

```
HP@DESKTOP-CV29E6P MINGW64 ~/Downloads/New Project (feature-registration)
$ git checkout master
Switched to branch 'master'
Your branch is up to date with 'origin/master'.
```

3.2 Merge `feature-login` into `master`

- Merge the `feature-login` branch into the `master` branch:

  `git merge feature-login`

```
HP@DESKTOP-CV29E6P MINGW64 ~/Downloads/New Project (master)
$ git merge feature-login
Updating 6a0b64f..a060d78
Fast-forward
 README.md | 1 +
 1 file changed, 1 insertion(+)
```

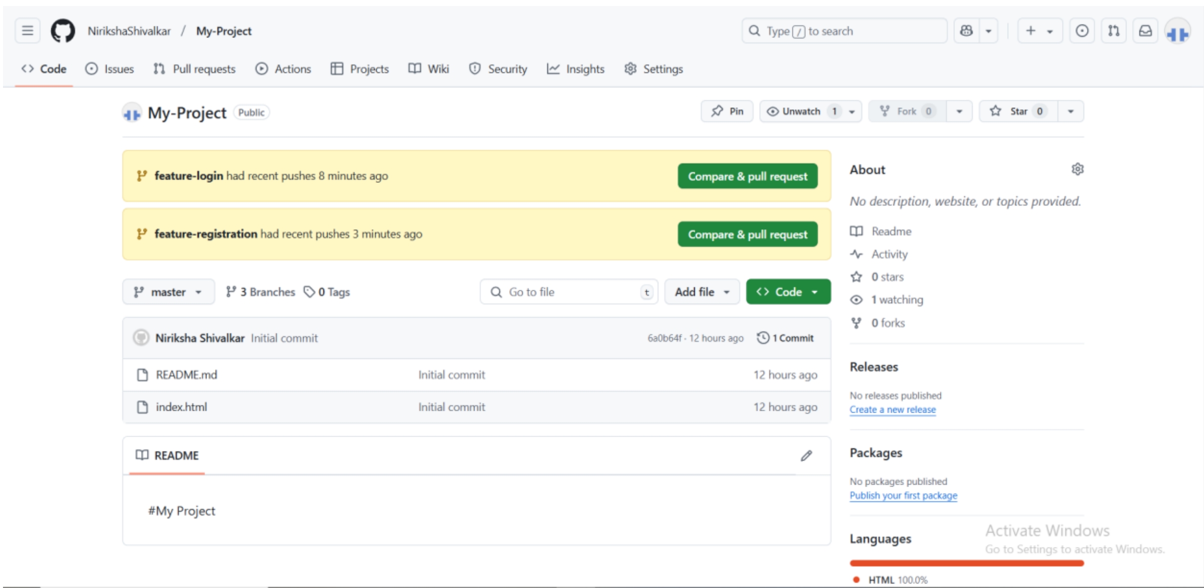
- Since the changes are in different parts of the file, the merge will succeed without conflict.
-

3.3 Merge `feature-registration` into `master`

- Now, try to merge the `feature-registration` branch:

  `git merge feature-registration`

```
HP@DESKTOP-CV29E6P MINGW64 ~/Downloads/New Project (master)
$ git merge feature-registration
Updating a060d78..8fe14f0
Fast-forward
 README.md | 1 +
 1 file changed, 1 insertion(+)
```



3.4 Resolve the Merge Conflict

- Open the `README.md` file. You will see something like this:

```
<<<<<<< HEAD
Login feature
=======
Registration feature
>>>>>>> feature-registration
```

- Edit the file to resolve the conflict. For example, you might combine both features like this:

```
markdown
Login feature
Registration feature
```

```
C: > Users > HP > Downloads > New Project > ⓘ REA
1    #My Project
2    Login feature
3    Registration Feature
4
```

3.5 Stage the Resolved Conflict

- Once the conflict is resolved, stage the file:
  `git add README.md`

```
HP@DESKTOP-CV29E6P MINGW64 ~/Downloads/New Project (master)
$ git add README.md
```

3.6 Commit the Merge

- Commit the merge with a message:
  `git commit -m "Resolve merge conflict between feature-login and feature-registration"`

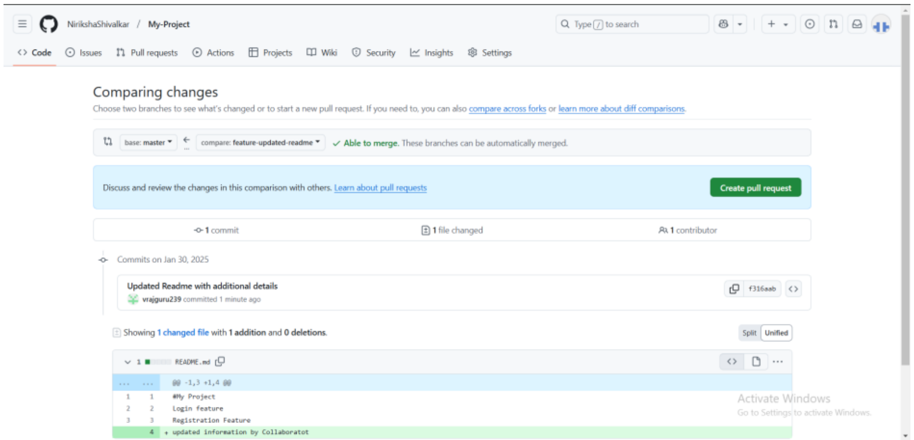
```
HP@DESKTOP-CV29E6P MINGW64 ~/Downloads/New Project (master)
$ git commit -m "Conflicts resolved"
On branch master
Your branch is ahead of 'origin/master' by 2 commits.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
```

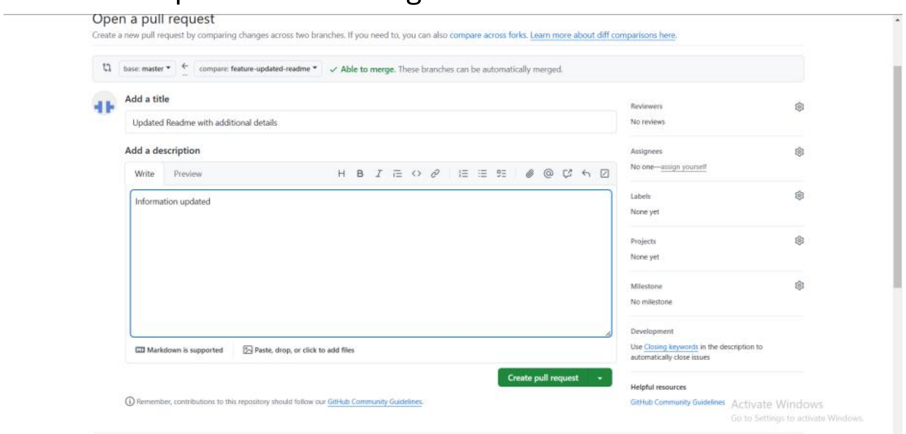
Step 4: Perform a Pull Request Review and Handle the Integration Process

4.1 Create a Pull Request (PR) on GitHub

- Go to your repository on GitHub.
- You should see a button to create a Pull Request for the branches you've pushed (`feature-login` and `feature-registration`).
- Click on "New Pull Request" and select the `feature-login` branch and compare it with `master`.

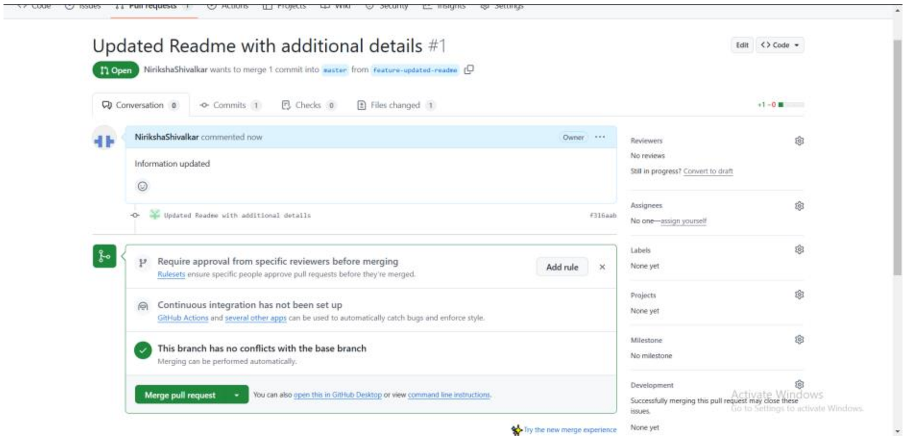


- After reviewing the changes, click "Create Pull Request."
- Add a description of the changes and create the PR.



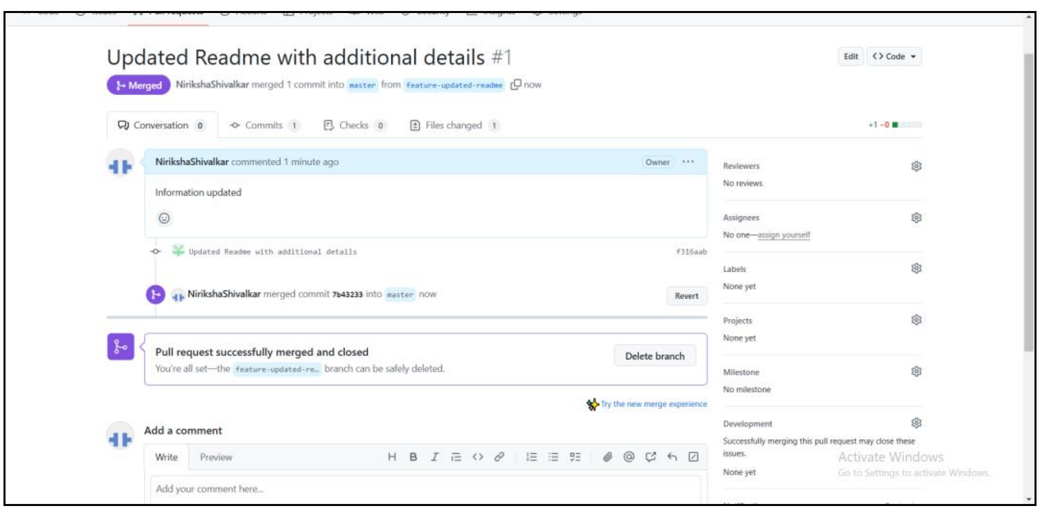
4.2 Review the Pull Request

- Review the changes in the pull request.
- You can see the changes made and leave comments on specific lines of code if necessary.
- You can ask a team member to review it as well or approve it yourself.



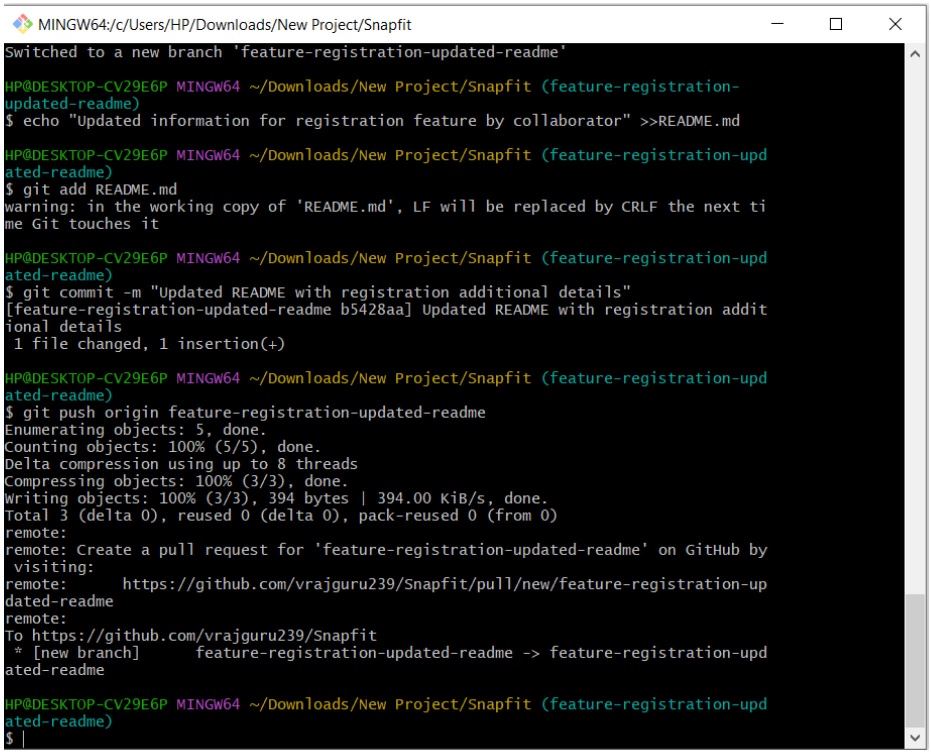
4.3 Merge the Pull Request

- After the PR review, click the "Merge pull request" button.
- Choose "Confirm merge" to integrate the changes from `feature-login` into `master`.

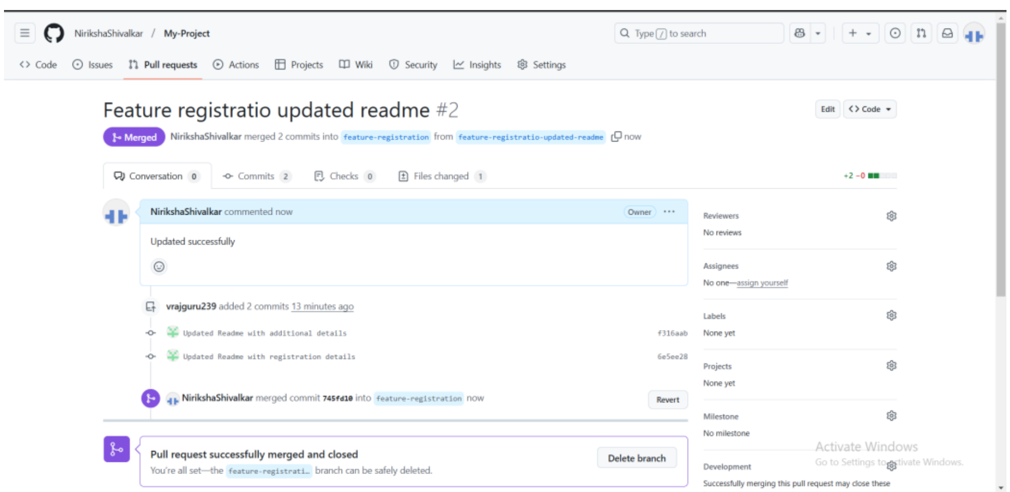


4.4 Repeat the Process for `feature-registration`

- Repeat the same process for the `feature-registration` branch.



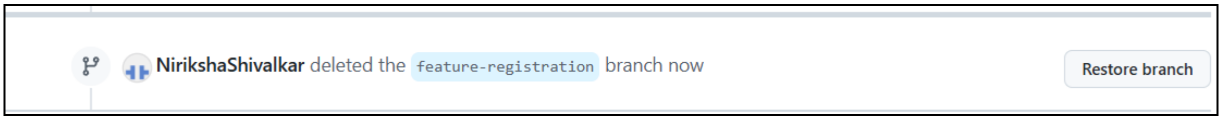
- Create a new PR for `feature-registration`, review it, and merge it into `master`.



4.5 Clean Up the Branches

- After successfully merging the feature branches, delete them from GitHub (optional):

```
git push origin --delete feature-login
git push origin --delete feature-registration
```



Step 5: Connect GitHub Cloud to Jira

These instructions are for connecting **GitHub Cloud** or **GitHub Enterprise Cloud** to Jira. Show me how to connect GitHub Enterprise Server

When you connect a GitHub Cloud organization to Jira, your team can link their development activity to Jira issues. This lets you track branches, commits, and pull requests in the context of your Jira issues, on your Jira board, in the releases feature, and more.

Before you begin

To install and set up the GitHub for Jira app, you need:

- Site administrator permission for your Jira site.
- Organization owner permission for your GitHub organization.

For some organizations, the task of connecting GitHub to Jira might involve multiple team members:

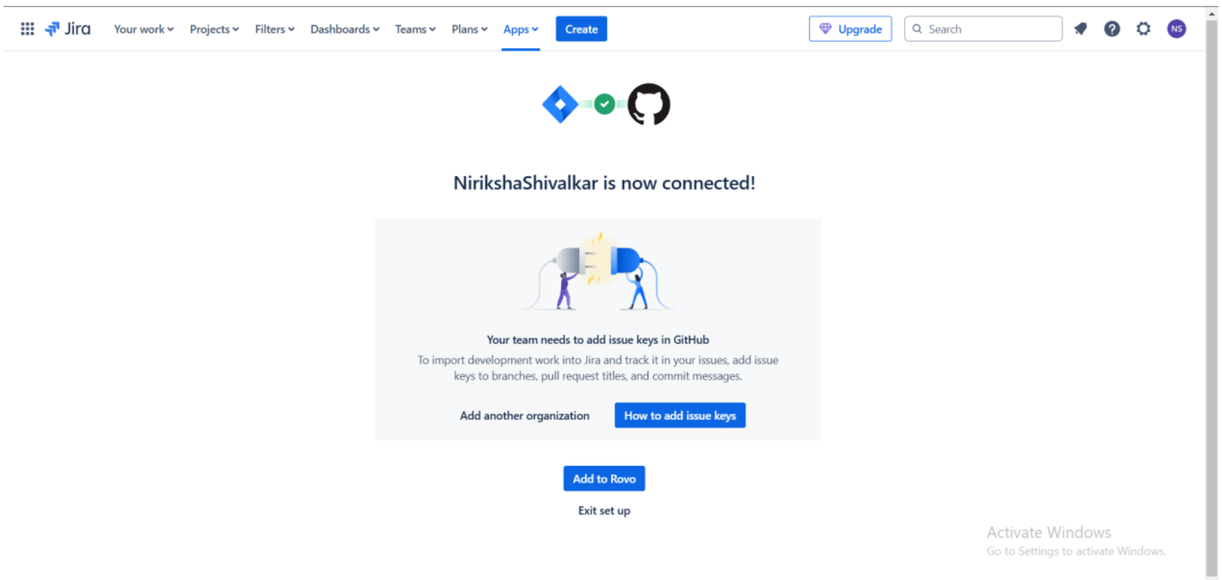
- A Jira site admin will install the GitHub for Jira app.
- A GitHub organization owner will connect a GitHub organization to your Jira site.

5.1 Install the GitHub for Jira app

1. In Jira, select **Apps**, then select **Explore more apps**.
2. Search for **GitHub for Jira**, then select it from the results.
3. Select **Get app**, then **Get it now**.

5.2 Connect a GitHub organization

1. After the app is installed, select **Get started**. If the app is already installed on your Jira site, you can find this section by selecting **Apps**, then **Manage your apps**, and then **GitHub for Jira**.
2. Select **Continue**.
3. Select **GitHub Cloud**, then **Next**.
4. Enter your GitHub username and password, then **Sign in**.
5. Find the organization you want to connect to Jira, then select **Connect**.

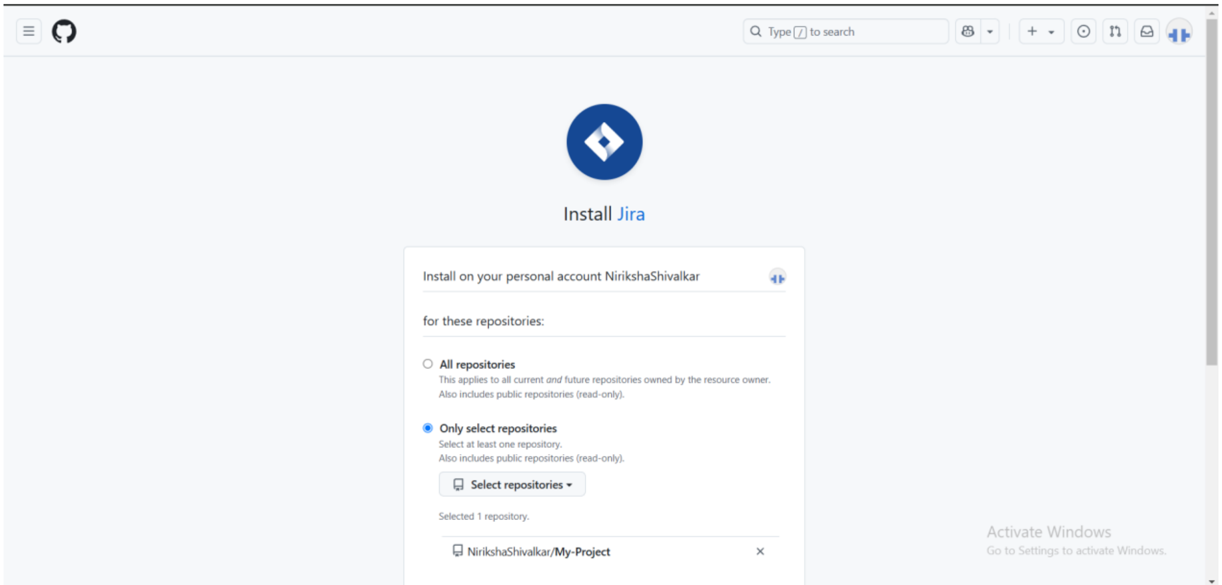


To check your permissions for a GitHub organization, open your GitHub organization settings and look for your permission level next to the organization name. Organization owners can review and accept permission requests from Jira in your organization settings. More about required permissions for GitHub for Jira

5.3 Add the Jira app to a new GitHub organization

If no organizations are available to connect to Jira, you'll need to install the Jira app on a new GitHub organization.

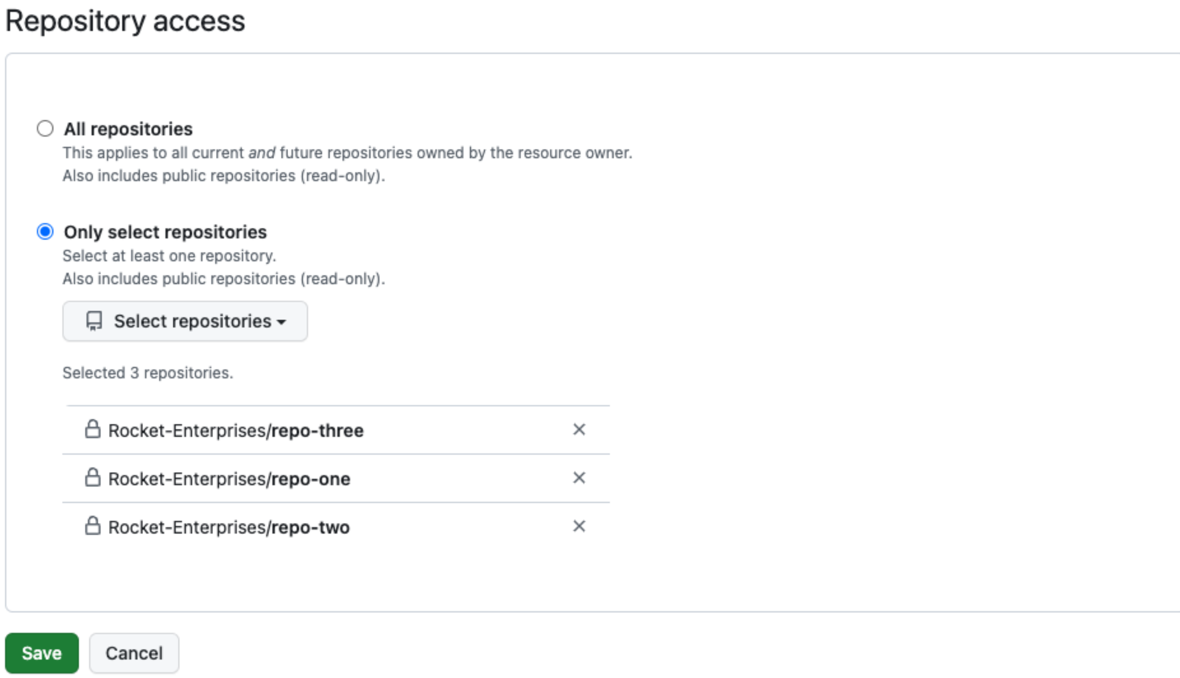
1. From step five in the instructions above, **Select an organization in GitHub.**
2. The GitHub site will open in a new tab and show a list of all the organizations available in your GitHub account. Select the organization you want to connect to Jira.
3. Choose the repositories you want to give Jira access to by selecting either **All repositories** or **Only select repositories**.
4. Select **Install.**



5.4 Connect a new GitHub repository

- If you selected **All repositories** when you connected a GitHub organization, Jira will be able to access all the repositories in that organization, including any new repositories you create.
- If you selected **Only select repositories**, any new repositories you create won't be added automatically. Here's how to add them:

1. In Jira, select **Apps**, then select **Manage apps**.
2. In the sidebar, under **GitHub for Jira**, select **Configure.**
3. Find the organization that contains the repo you want to add and select the three-dot menu (**...**) to view available actions.
4. Select **Configure**.
5. A new tab will open with your GitHub organization settings. Under **Repository access**, select the dropdown **Select repositories**.
6. Select the repository you want to connect to, then **Save.**



If you use IP allowlists in your GitHub organization, you might have issues using GitHub for Jira, even if the correct IP addresses are in your IP allowlist. Here's the workaround: How to update your GitHub IP allowlist configuration

Still need help? Raise an issue with our team.

5.5 Link your development information to Jira issues

To link branches, commits, and pull requests to Jira, your team must include Jira issue keys in their development actions.

1. Find the issue key for the Jira issue you want to link to, for example "JRA-123". You can find the key in several places in Jira:
   - On the board, issue keys appear at the bottom of a card.
   - On the issue's details, issue keys appear in the navigation at the top of the page.
2. Check out a new branch in your repo, using the issue key in the branch name. For example, `git checkout -b JRA-123-<branch-name>`.
3. When committing changes to your branch, use the issue key in your commit message to link those commits to the development panel in your Jira issue.
   For example, `git commit -m "JRA-123 <summary of commit>"`.
4. When you create a pull request, use the issue key in the pull request title.

- After you push your branch, you'll see development information in your Jira issue.

```
HP@DESKTOP-CV29E6P MINGW64 ~/Downloads/New Project (feature-login-updated)
$ git commit -m "TMA-1:Login issue fixed"
[feature-login-updated 557071c] TMA-1:Login issue fixed
 2 files changed, 2 insertions(+)
 create mode 160000 My-Project
 create mode 160000 Snapfit

HP@DESKTOP-CV29E6P MINGW64 ~/Downloads/New Project (feature-login-updated)
$ |
```

How to use the GitHub for Jira app

- Link GitHub development information to Jira issues
- Link GitHub workflows and deployments to Jira

Post lab exercise:

Instructions:

1. Initialize a Git repository and push it to GitHub.
2. Create and merge multiple feature branches.
3. Simulate a merge conflict and resolve it.
4. Perform a pull request review and handle the integration process.

1. Initial Setup:

Each developer should clone the repository and create a feature branch as instructed.

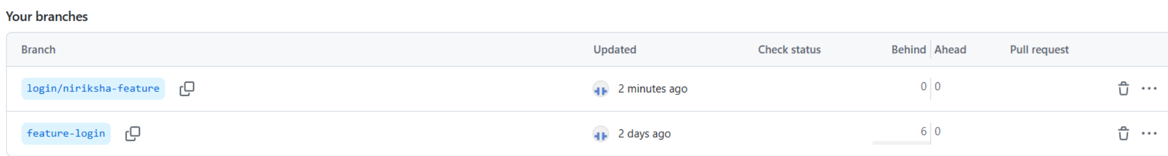
```
# Clone the repository
git clone https://github.com/TechCo/your-repo.git

# Navigate into the project directory
cd your-repo

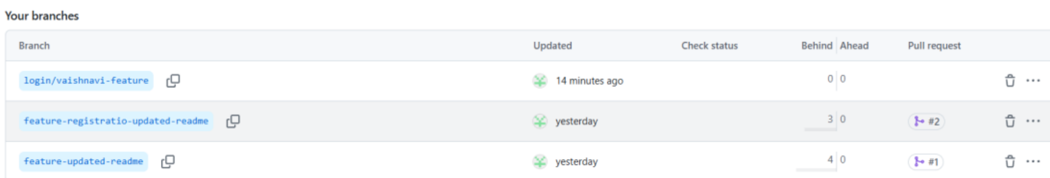
# Create a feature branch
git checkout -b feature/john-does-feature
```

Developer 1 and **Developer 2** should create separate feature branches based on the above instructions. For example:

- Developer 1 creates the branch: `feature/john-feature`

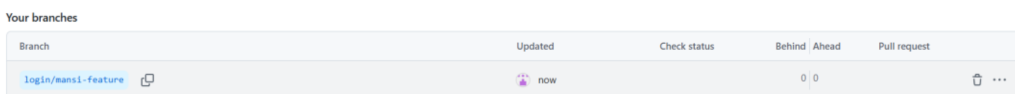


- Developer 2 creates the branch: `feature/jane-feature`



-

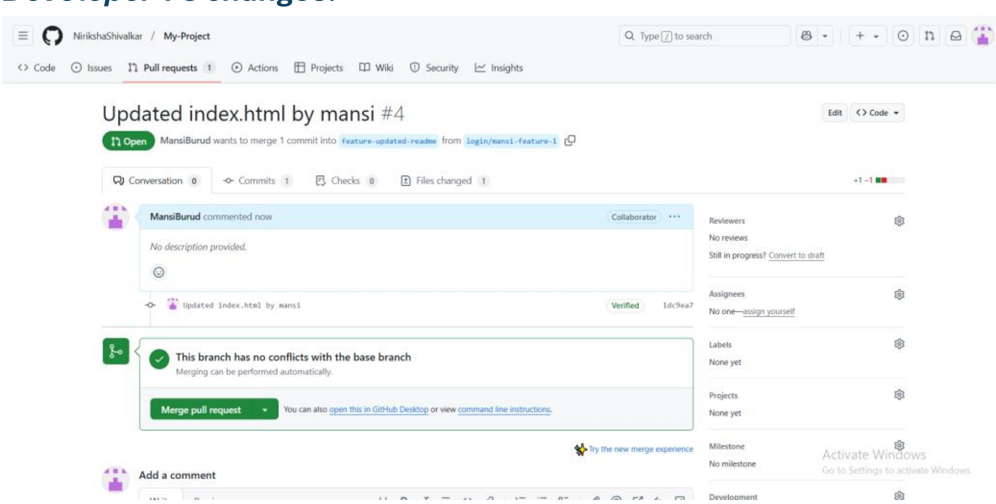
Developer3 creates the branch



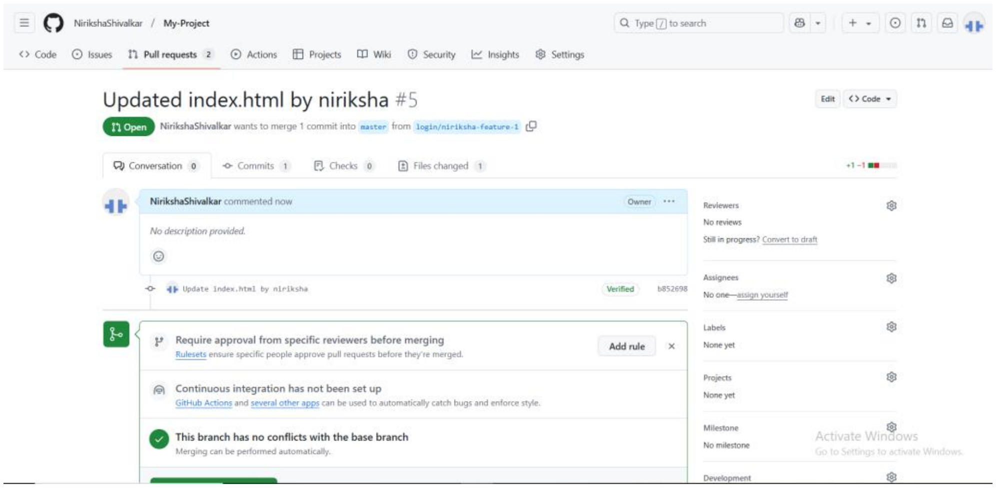
2. Collaborative Changes:

Both developers will make changes to the same file (`app.js`), but in different parts of the code.

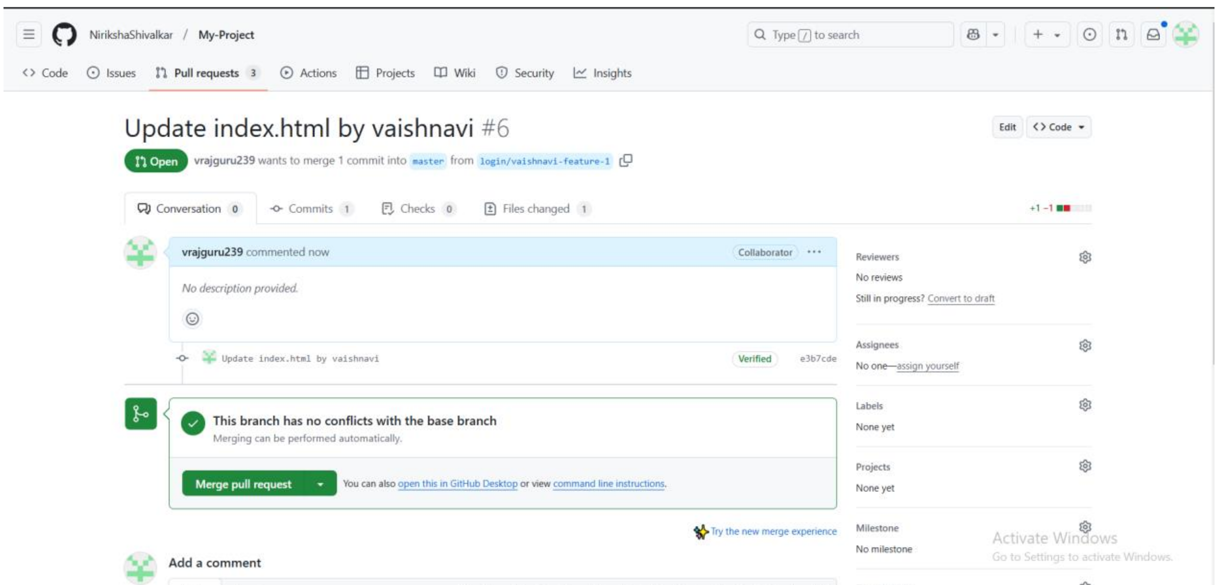
Developer 1's changes:



Developer 2's changes (on the same line):



Developer 3’s changes



3. Merge Conflict:

Once both developers have pushed their changes, it’s time to merge their feature branches into `main`. This will result in a merge conflict since both developers modified the same lines of code in `app.js`.

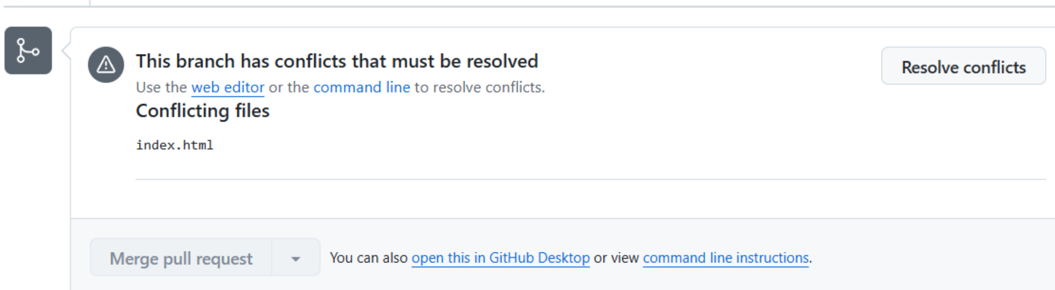
Steps for Merging and Resolving the Conflict:

- Switch to the `main` branch:

```
git checkout main
git pull origin main  # Fetch the latest changes from the main branch
```

- Try merging one of the feature branches into `main` (e.g., Developer 1’s branch):

```
git merge feature/john-feature
```



- At this point, Git will notify that a conflict exists in `app.js`. The conflicted file will be marked as having conflicts. To resolve it:

1. Open `app.js` in a code editor. You will see something like this:

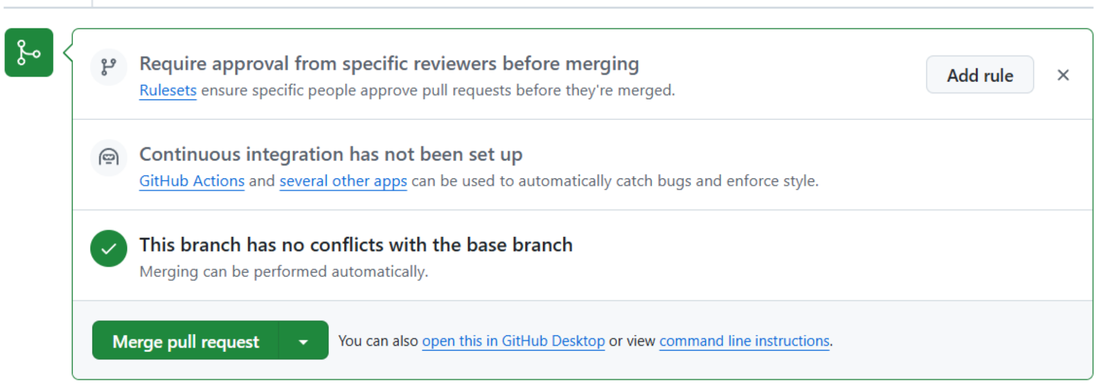
```
function greetUser() {
    console.log("Hello, user!");
<<<<<<< HEAD
    console.log("Welcome to TechCo's new app.");
=======
    console.log("We hope you enjoy using TechCo's app.");
>>>>>>> feature/jane-feature
}
```

2. The conflict markers (<<<<<<<, =======, >>>>>>>) indicate where the conflict occurs. You need to decide which version to keep or how to merge the changes. Here’s how you can resolve it:

```
function greetUser() {
    console.log("Hello, user!");
    console.log("Welcome to TechCo's new app.");
    console.log("We hope you enjoy using TechCo's app.");
}
```

3. After resolving the conflict, remove the conflict markers and save the file.

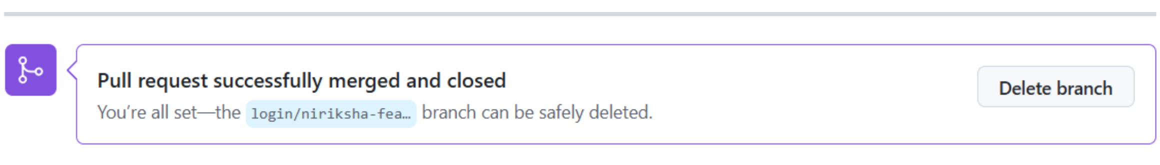
- Add the resolved file and commit the changes:
- ```
  git add app.js
  git commit -m "Resolve merge conflict in app.js"
  git push origin main
  ```

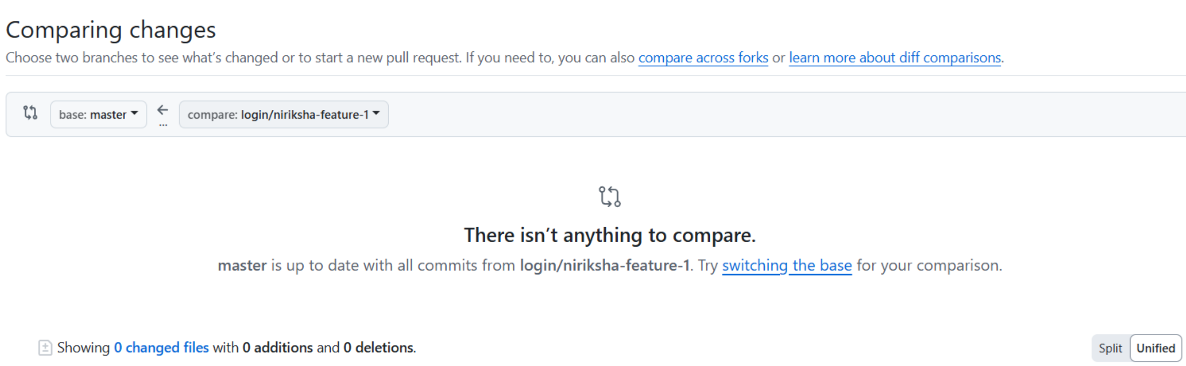


4. Rebasing Practice:

Now, Developer 1 and Developer 2 will fetch the latest changes from the `main` branch and rebase their feature branches onto `main` to ensure that their code is up-to-date.

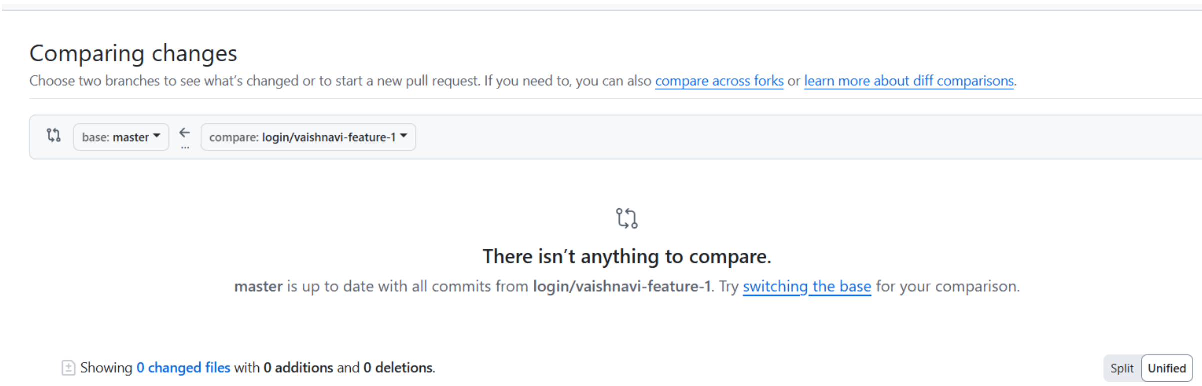
Developer 1 Rebasing:



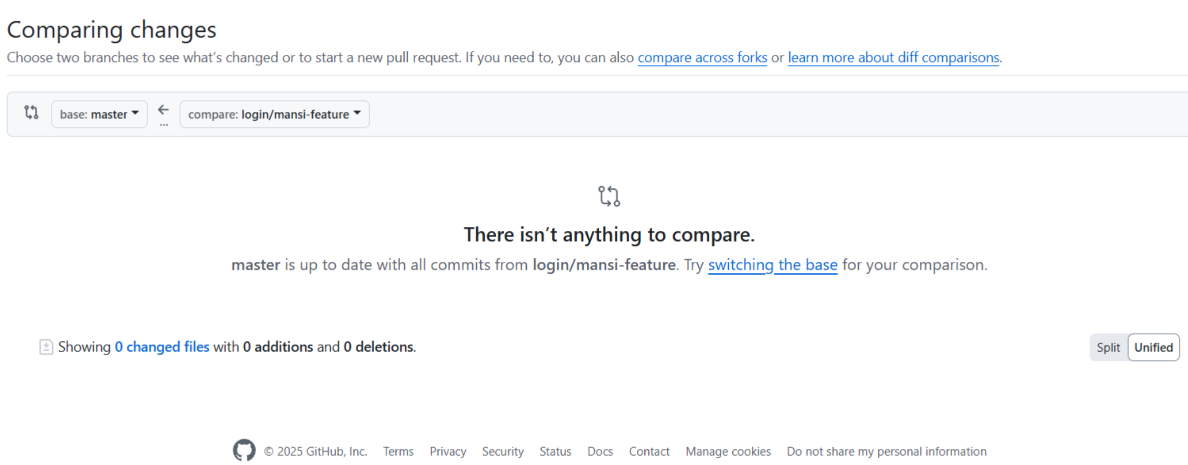

  ```

Developer 2 Rebasing:

Developer 2 will repeat the same steps to rebase their branch.



`Developer 3:`



5. Reviewing Commit History:

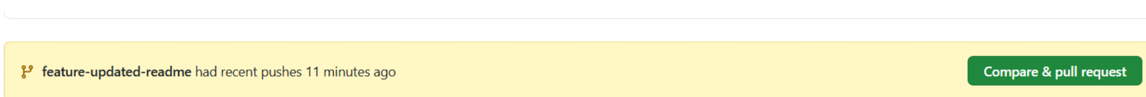
Once the merge and rebasing are complete, both developers should inspect the commit history to ensure that their commits are clear and meaningful.

```
git log --oneline --graph
```

Ensure that each commit message follows a clear and consistent format, e.g., "Fix bug in the greeting function" or "Add new user welcome message."

6. Final Task: Creating Pull Requests (PRs):

- Developer 1 and Developer 2 should now open pull requests on GitHub to merge their feature branches into the `main` branch.



Conclusion: By following these steps, the developers will:

- Learn how to resolve merge conflicts effectively.
- Practice using rebase to keep their branches up to date with `main`.
- Understand how to maintain clean commit histories and write meaningful commit messages.
- Gain hands-on experience with Git workflows that are crucial for collaborative development.